WCS / BRASIL

INAUGURAL EDITION · SEPTEMBER/OCTOBER 2026

Monthly evolution report

Illegal Wildlife Trafficking Global Observatory

21 September — 15 October 2026

Conservation Intelligence Program | WCS Brasil

SCOPE AND TRACEABILITY NOTE

The period starts on 21 September, but software and architecture development had begun at least two weeks earlier.

15 October 2026

ENGLISH

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ENGLISH · EXECUTIVE SUMMARY

Executive summary

Between 21 September and 15 October 2026, the Illegal Wildlife Trafficking Global Observatory evolved from a seizure-mapping application into a conservation intelligence infrastructure. This transition brought together a public dataset of events and evidenced routes, four categories of contextual references, DOI-linked reports, complementary geospatial visualizations, use analytics, and a restricted workflow for wildlife-market monitoring.

The dataset expanded sharply: approximately 92 events at the initial checkpoint compared with 1,047 in the reference canonical build of 15 October; documented routes increased from 17 to 274. At the same cutoff, the scientific repository held 125 papers, the journalism repository 172 investigative and analytical stories, and the institutional collection 15 documents. Four weekly editions were catalogued.

Methodological development was equally important. The map was decoupled from the underlying structured data; standardized fields addressed quantities and geographic precision; new entries required deduplication, traceable evidence, case-specific contextual analysis, and editorial parity in Portuguese, English and Spanish. Rules were introduced to distinguish evidenced movements from attributed travel and to prevent incompatible animal, egg, product and weight counts from being combined.

The restricted market-intelligence area, initiated in October, extended the mission beyond seizures to listings, physical vendors, markets and trading platforms. At 15 October, not every private-data import or synchronization was complete; accordingly, this report distinguishes verified implementations, work still being stabilized and planned capabilities.

The immediate strategic priorities are integration with Aurum — including official open-data seizures and cases investigated by WCS Brasil that often receive no media coverage — an additional protection layer for intelligence reports, and completion of the journal issue devoted to wildlife trafficking in Latin America.

Metric	15 Oct reference
Events	~92 → 1,047
Routes	17 → 274
Science [S]	125 papers
Journalism [N]	172 stories
Institutional [I]	15 documents
Weekly editions	4 reports

01 Background and reporting scope

The Observatory is an analytical and operational resource developed within WCS Brasil's Conservation Intelligence Program. It seeks to structure evidence on illegal wildlife trade, make documented movements understandable geographically, facilitate source review and support scientific and institutional reporting.

The start of this chronology on 21 September is a documentation checkpoint rather than the conception date. By then, code, architectural decisions, map prototypes and datasets had already been under development for

at least two weeks. The statement that the platform was “created on 21 September” would therefore be inaccurate.

Counts in this report are historical checkpoints recovered from commits and catalogues, not a reconstructed count for every individual day. The date a record enters the Observatory differs from the date of the wildlife incident. Terms such as “about”, “by” and “in one build” indicate limited precision.

SCOPE AND TRACEABILITY NOTE

This is the first monthly report on the Observatory’s technological and methodological evolution. The documented reporting window runs from 21 September through 15 October 2026: 25 calendar days, not the complete age of the project. Software code, interface prototypes and the data architecture had already been under development for at least two weeks before 21 September, reaching back to at least early September 2026. As commits preceding the reporting baseline have not been fully reconstructed, this report does not assign an exact project inception date.

02 Development chronology

Phase 1 | 21–27 September — foundations and canonical datasets

21 Sep | The MAP 2D and GLOBE 3D modes were already available, together with event cards, bottom dashboard, language selection and an initial layer of 17 documented routes spanning 21 segments. Inputs for weekly reports were prepared, including retrospective coverage of 8–14 September.

22 Sep | The data and presentation layers were separated: cases.json, routes.json and metadata.json were consolidated on GitHub. Dataset reconciliation and 18 validated Brazilian cases brought the reference set to 110 cases. Structured papers.json [S] and news.json [N] repositories were established with multilingual metadata and discovery pipelines.

23–24 Sep | Coverage expanded across Africa, Europe, the Americas, Asia and Oceania, including invertebrate-focused batches. Source photographs were reviewed and additional routes added only when justified. PT/EN/ES text search and country, period and taxon filters became increasingly structured.

25–26 Sep | Three Amazon batches added 21 events and eight associated routes, alongside additional coverage of China, Russia, India, Pakistan, Arab countries and under-represented geographies. Route-feed synchronization was corrected; approximately 85 routes were referenced during this stage.

27 Sep | Event storage began shifting to individual records compiled into a single public feed. The Reports section was reorganized into weekly, semiannual and annual catalogues. The dataset had reached approximately 390 events by this stage.

Phase 2 | 28 September–5 October — editorial operations and thematic expansion

28–30 Sep | The 15–21 September weekly report was published and catalogued. Ingestion became more granular and frequent, with event date, publication date and ingestion date kept distinct. By 30 September, development primarily favored GitHub and the existing infrastructure.

1 Oct | Cases began supporting thematic analyses on parrots, eggs, Amazonian turtles, sharks and Mato Grosso do Sul. A seizure-related study from that Brazilian state was added to the scientific repository. The “How it works” page was already confirmed by this period, without a verifiable original deployment date.

2 Oct | A separate Publications section was created for analytical texts, journal articles and other technical outputs, distinct from Reports, Scientific research and Journalism.

3 Oct | Google Analytics 4 instrumentation was integrated to examine usage and interactions: opening event cards, switching between MAP/GLOBE, filtering, language changes and access to publications.

4–5 Oct | Historical case sets and thematic collections expanded. International geometries, routes and coordinate precision were refined, reinforcing that an ingestion date is not the date of the underlying incident.

Phase 3 | 6–10 October — contextual intelligence, transparency and scale

6 Oct | The news repository [N] expanded with investigations about Vantara, transnational networks, bird-egg trafficking and Southeast Asian markets, together with corrections to source metadata and dates.

7–8 Oct | Map route rendering was refined with grey lines, blue directional arrows, trilingual labels, clickable evidence and explicit separation of documented journeys, attributed movements and broader network context. The dataset exceeded 700 events.

9 Oct | The recent-event highlight moved to a roughly 48-hour window based on occurred_at, applying conservative rules when the hour was unknown. The institutional catalogue [I] launched with 15 documents in reports-institutional.json. The 29 September–5 October weekly edition was published.

10 Oct | Thirty additional scientific papers and fifteen analytical stories were incorporated. Institutional citation guidance and the testing-stage notice were standardized. A synchronized checkpoint contained 846 events and 213 routes.

Phase 4 | 11–15 October — restricted intelligence and dashboard maturity

11 Oct | The Wildlife Market Intelligence Hub (NIMF) was structured, using a private repository and Firebase/Firestore access architecture. Four monitoring categories were defined: listings, physical vendors, markets and platforms.

12 Oct | Restricted triage, layer-management and records panels were developed, alongside MAP/GLOBE controls, light/dark themes and source and coordinate audits for wildlife markets. Prepared items did not necessarily equal points visible in the published map.

13 Oct | The public dashboard was revised: one horizontally stacked bar by faunal group replaced less effective representations, ambiguous "+ ?" values were removed and a day-level cumulative event chart was added from 21 September onward.

14 Oct | Public filters and navigation functionality were restored. A historical batch of 54 shark and ray product events and 36 sourced routes was published; one build recorded 1,022 events.

15 Oct | The restricted area advanced through *Ceratophrys ornata* listing imports and corrections to names, currencies and biological categories. Trilingual [O/S/N/I] contexts were audited, weak analogies removed, new cases added and the red recent-event markers fixed. Final reference: 1,047 events, 274 routes, 125 [S] papers and 172 [N] stories.

03 Data architecture and integration

The wildlife-trafficking-data repository established canonical files independently of the web interface. cases.json is the compiled public event feed; routes.json contains evidenced routes; and metadata.json specifies source and feed metadata. Individual event records were progressively organized under cases/YYYY/MM/ and validated before being compiled and published.

Structured fields address stable event IDs, incident and publication dates, coordinates and geographic precision, taxonomic groups, quantities and qualifiers, original sources, validation status, PT/EN/ES cards and contexts, and rules for aggregate operations or records excluded from totals. Mass, volume and packaging are not converted into counts of animals.

Routes must be linked to a specific incident, backed by a documentary source and described with credible endpoints, direction, precision and trilingual evidence. Unsubstantiated movements are excluded; broad network context must remain separate from observed travel.

Record-level storage improved maintenance, but the public compiled cases.json was still a single payload. Public-feed partitioning, periodic archival and the proposed temporary.json/cleaning.json workflows were under discussion and not fully operational by the reporting cutoff.

04 Repositories, documents and publications

Four evidence categories became distinct components: [O] primary official or media sources tied to incidents; [S] peer-reviewed and other scientific literature with bibliographic metadata; [N] substantial investigative reporting on networks and markets; and [I] institutional reports and assessments. Secondary references are attached only when they materially inform the specific case.

Reports were arranged into weekly, semiannual and annual catalogues, with PDFs deposited in Zenodo and persistent DOIs. The separate Publications section, introduced on 2 October, serves research papers and other technical outputs. The Observatory's institutional DOI is 10.5281/zenodo.23282620.

At the 15 October cutoff, four weekly editions were catalogued; the annual and semiannual JSON catalogues existed but contained no entries in the inspected files. That observation does not rule out reports drafted or distributed outside the dynamic catalogue.

Repository / file	Milestone	15 Oct status
[O] Events	22 Sep — canonical data	1,047 events; cases.json
Documented routes	21–22 Sep — dynamic layer	274 routes; routes.json
[S] Scientific research	22 Sep — papers.json	125 papers
[N] Investigative reporting	22 Sep — news.json	172 stories
Weekly reports	21 Sep first files; 27 Sep catalogue	4 editions
Semiannual and annual	27 Sep — catalogues	Files exist; no entries in inspected JSON
Publications	2 Oct — separate section	Scientific and technical outputs
[I] Institutional reports	9 Oct — reports-institutional.json	15 documents
Restricted area / NIMF	11 Oct — first architecture	4 categories; synchronization evolving

Catalogued weekly editions

Coverage	Publication	DOI
8–14 Sep	27 Sep	10.5281/zenodo.22997274

Coverage	Publication	DOI
15–21 Sep	28 Sep	10.5281/zenodo.23008516
22–28 Sep	2 Oct*	10.5281/zenodo.23100929
29 Sep–5 Oct	9 Oct	10.5281/zenodo.23269208

* The 22–28 September edition was entered into the catalogue on 6 October; its published_at field reads 2 October. The next edition, 6–12 October, was not yet published at cutoff.

05 Visualizations, metrics and user experience

MAP 2D provides a conventional cartographic view; GLOBE 3D offers global visualization, later refined so the far hemisphere does not project its markers onto the visible surface. Both modes consume shared datasets and permit linked event selection. The route layer shows directional geometry and evidence on selection.

Users can refine results by period, country, taxon and free-text queries across Portuguese, English and Spanish. Light/dark theme control, fullscreen viewing, detail panels and CSV export were developed; the 14 October restoration ensured map filters continued to work after a redesign.

The bottom dashboard shifted toward a single horizontally stacked taxonomic bar and a daily cumulative insertion chart beginning 21 September. Recent events use a rolling approximately 48-hour red marker rule, with an important behavioral correction completed on 15 October.

Google Analytics 4 instrumentation began on 3 October to differentiate meaningful feature use from raw visit totals. Instrumentation was deployed, but some event-delivery and interpretation checks remained open.

Feature	Development checkpoint
MAP 2D / GLOBE 3D	Already available 21 Sep; canonical integration 22 Sep
Routes, evidence and direction	Since 21 Sep; refinements on 7–8 Oct
PT/EN/ES search and filters	Consolidated 24 Sep; filters restored 14 Oct
Quantitative dashboard	Structured fields 22 Sep; redesign 13 Oct
How it works page	Confirmed available by 1 Oct
Publications and reports	Publications 2 Oct; institutional catalogue 9 Oct
Google Analytics 4	Instrumentation 3 Oct
Taxon bar + cumulative series	13 Oct
Red recent-event markers ~48h	Rule on 9 Oct; correction on 15 Oct
NIMF and restricted area	From 11 Oct

06 Restricted area and market intelligence

The NIMF restricted environment targets wildlife commercialization intelligence rather than only seizure response. It categorizes listings, physical vendors, markets and platforms, retaining each record's evidence type and avoiding unsupported claims about legality.

Its architecture combines a private GitHub repository, intake validation, Firebase/Firestore authentication and a specialized map. Prepared records, merged code, synchronized Firestore data and visible map markers are different publication states and require independent verification.

From 12 to 15 October, the team worked on market-source validation, coordinates, thematic layers and species-specific advertisements. *Ceratophrys ornata* records highlighted the importance of scientific-name italics, original listing currency and correct biological categories; frogs must not be classified as "colonies".

The present restricted environment is not yet the additional security layer planned for operational intelligence reports. That future module requires dedicated access rights, document-level separation, user roles and access traceability.

07 Methods, evidence and quality assurance

The intake method matured into sequential steps: multilingual global discovery; official sources prioritized and journalistic corroboration; confirmation of incident date, location, taxon, alleged offence and evidence quality; deduplication against cases.json; case-specific contextual analysis; route assessment; schema validation; and publication.

Uncertainty must be preserved. Trilingual content was audited for language contamination, superficial academic parallels and unsupported claims. [S], [N] and [I] references are not inserted for appearance; they must clarify relevant mechanisms, markets, law or documented patterns.

Quantity assessment distinguishes exact, minimum, approximate and unknown counts, as well as animals, eggs, specimens, wildlife products and biomass. Aggregate operations and duplicate news coverage must not inflate totals.

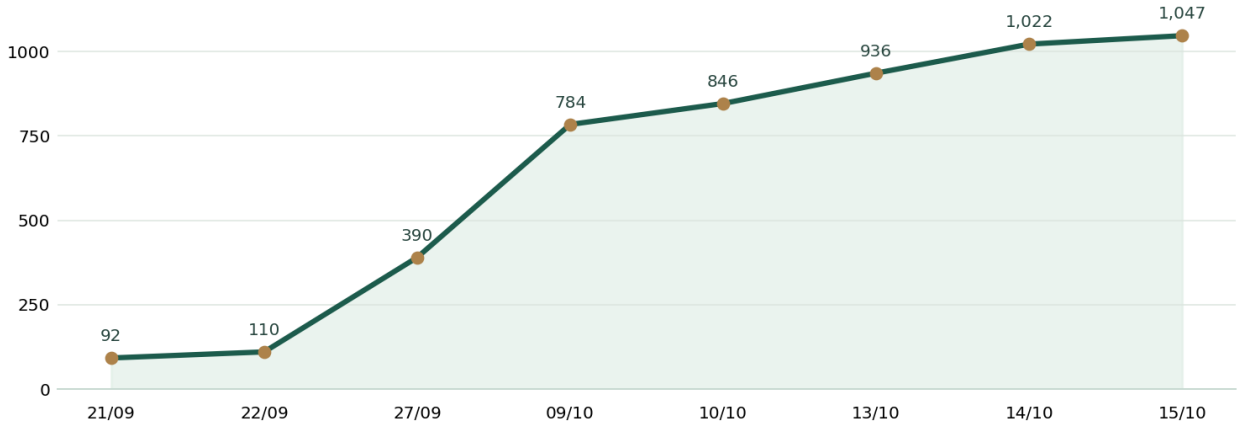
For routes, capture origin, shipment point, intended destination and seizure location are not interchangeable. Coordinate precision must follow the source rather than turn a general municipality into an artificially exact point.

08 Quantitative results and deliverables

Recorded incidents increased approximately 11.4-fold between the initial checkpoint and 15 October, while evidenced routes increased from 17 to 274. These are records ingested by a source-dependent system: they do not measure total worldwide trafficking and are not a statistically representative census of wildlife crime.

Other outputs go beyond the counts: editorial catalogues, DOI-linked publications, trilingual content, evidence audits, 2D/3D geospatial maps, searchable filters, taxon-level metrics, time-sensitive displays and the restricted market-intelligence workflow.

Figure 1. Cumulative growth in recorded events at selected checkpoints, 21 September–15 October 2026.



Documented checkpoints; approximate values where indicated. This is not a complete daily time series.

Checkpoint	Events
21/09	92
22/09	110
27/09	390
09/10	784
10/10	846
13/10	936
14/10	1,022
15/10	1,047

09 Limitations and technical backlog

Outstanding concerns include continued growth of compiled cases.json, consistency between canonical commits and the published feed, geographic and route precision, synchronization lag, systematic PT/EN/ES contextual review, access documentation and secure operations for private market intelligence.

The proposed temporary intake and cleaning.json audit trail, time-based archival and monthly partitioning remained technical plans rather than fully deployed systems. Similarly, proposed or scheduled searches cannot be described as successfully autonomous without corresponding execution and publication logs.

Counts are snapshots from builds at different moments; intermediate variation may reflect additions, corrections, deduplication and timing differences. The checkpoint table is not a substitute for a complete day-by-day transaction history.

10 Strategic next steps

Priority 1 — Integrate Aurum data

Build a controlled connection to Aurum, the WCS Brasil intelligence platform that integrates seizures found in official open datasets and events directly investigated by the team, including numerous cases never reported by the media. Address schema harmonization, provenance, cross-system deduplication, permissions and the separation of public versus restricted information. This will reduce reliance on media visibility alone.

Priority 2 — Intelligence reports behind a second protection layer

Develop an intelligence-report module accessible only to specifically authorized users, with a security layer additional to the current restricted-area login. Implement role- and document-level authorization, content segregation, audit logs, retention and sharing rules, and mandatory human review before sensitive release.

Priority 3 — Complete the Latin America wildlife-trafficking journal issue

Finalize editorial production of the journal issue focused on wildlife trafficking in Latin America: manuscript review, standardization of figures and tables, reference checks, consistent terminology, version tracking, metadata and dissemination. Observatory and Aurum evidence may support analyses where authorized and methodologically justified.

Cross-cutting technical work

Complete data and route audits, monitor temporal-marker behavior, stabilize the restricted environment, verify Analytics events and decide on data partitioning, archival and cleanup before increasing ingestion frequency.

11 Conclusions

Within 25 documented days, preceded by at least two weeks of prior software and architectural development, the Observatory advanced from mapped events to an integrated infrastructure of data, evidence and analytical products. Quantitative expansion was accompanied by methodological safeguards, transparency, language accessibility and functional specialization.

The next stage should be judged less by the number of new points on a map and more by interoperability, access protection and actionable intelligence: integrating Aurum, safeguarding sensitive analytical reports and consolidating scientific publishing on wildlife trafficking in Latin America.

12 References and audit trail

Verification sources include canonical GitHub files and commits, report catalogues, Zenodo DOIs and interface-development records. Commit dates demonstrate data and code changes; by themselves, they do not verify rendered interface states or private synchronization.

Canonical events and routes / Ocorrências e rotas / Eventos y rutas — https://github.com/acarvalho-wcs/wildlife-trafficking-data

Scientific repository / Repositório científico / Repositorio científico — https://github.com/acarvalho-wcs/wildlife-trafficking-research

Journalistic repository / Repositório jornalístico / Repositorio periodístico — https://github.com/acarvalho-wcs/wildlife-trafficking-news

Reports catalogue / Catálogo de relatórios / Catálogo de informes — https://github.com/acarvalho-wcs/wildlife-trafficking-reports

NIMF — restricted GitHub repository / repositório restrito / repositorio restringido — https://github.com/acarvalho-wcs/wildlife-market-intelligence

Observatory institutional DOI / DOI institucional / DOI institucional — https://doi.org/10.5281/zenodo.23282620

Recommended institutional citation: Wildlife Conservation Society Brasil. (2026). Illegal Wildlife Trafficking Global Observatory. WCS Brasil. DOI: 10.5281/zenodo.23282620.

Editorial cutoff: 15 October 2026. Counts are historical checkpoints and may be revised through later canonical audits.